

Parity selection by spin- z jump operators acting on $|++\rangle$

Abstract

We consider a two-qubit system subjected to quantum jumps generated by the two collapseing operators $\hat{C}_\pm \propto S_z^{(1)} \pm S_z^{(2)}$, with the physical convention $\sigma_z |1\rangle = +|1\rangle$, $\sigma_z |0\rangle = -|0\rangle$. Starting from the product state $|++\rangle$ (the $+1$ eigenstate of σ_x on each qubit), we show that the first jump projects the system onto a definite z -parity subspace. A “positive” click (C_+) sends $|++\rangle$ to $-|\phi_-\rangle$ and then cycles deterministically $-|\phi_-\rangle \rightarrow |\phi_+\rangle \rightarrow -|\phi_-\rangle \rightarrow \dots$. A “negative” click (C_-) sends $|++\rangle$ to $-|\psi_-\rangle$ and then cycles $-|\psi_-\rangle \rightarrow |\psi_+\rangle \rightarrow -|\psi_-\rangle \rightarrow \dots$. This selection rule follows from the explicit commutation $[C_\pm, \Pi_z] = 0$ with the z -parity operator $\Pi_z = \sigma_z^{(1)} \otimes \sigma_z^{(2)}$. We also derive the jump probabilities at each step. Finally we analyze both concurrence $\mathcal{C}$ and spin squeezing parameter ξ^2 and derive ensemble averages.

1 Conventions, basis and Bell states

We work with two qubits in the computational z -basis $\{|1\rangle \equiv |\uparrow_z\rangle, |0\rangle \equiv |\downarrow_z\rangle\}$, with Pauli matrices $\sigma_\alpha^{(i)}$ acting on qubit $i = 1, 2$ and $\sigma_z |1\rangle = +|1\rangle$, $\sigma_z |0\rangle = -|0\rangle$. Spin-1/2 operators are $S_\alpha^{(i)} = \frac{1}{2}\sigma_\alpha^{(i)}$. The single-qubit x -eigenstates are

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

so the initial state is the product

$$|++\rangle = |+\rangle \otimes |+\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle).$$

We use the standard Bell states

$$|\phi_\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}} \quad (\text{even } z\text{-parity}), \quad (1)$$

$$|\psi_\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}} \quad (\text{odd } z\text{-parity}). \quad (2)$$

Define the z -parity operator

$$\Pi_z = \sigma_z^{(1)} \otimes \sigma_z^{(2)}.$$

Its eigenvalues are $+1$ on the even subspace (spin aligned), $\text{span}\{|00\rangle, |11\rangle\} = \text{span}\{|\phi_+\rangle, |\phi_-\rangle\}$ and -1 on the odd subspace (spin unaligned), $\text{span}\{|01\rangle, |10\rangle\} = \text{span}\{|\psi_+\rangle, |\psi_-\rangle\}$.

2 Quantum jumps dynamics

2.1 Collapse (jump) operators and commutation with parity

By adding a 50/50 BS the two collapsing operators are the balanced sum/difference of local z -spins with an overall rate factor:

$$\hat{C}_\pm = \sqrt{\gamma} \frac{S_z^{(1)} \pm S_z^{(2)}}{\sqrt{2}} = \sqrt{\gamma} \frac{\sigma_z^{(1)} \pm \sigma_z^{(2)}}{2\sqrt{2}}. \quad (3)$$

Any global prefactor only affects rates, not the direction of the post-jump state.

Since $\sigma_z^{(1)}$ and $\sigma_z^{(2)}$ commute and $(\sigma_z^{(i)})^2 = \mathbb{I}$,

$$[\sigma_z^{(1)}, \Pi_z] = \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(1)} \sigma_z^{(2)} = 0, \quad (4)$$

$$[\sigma_z^{(2)}, \Pi_z] = \sigma_z^{(2)} \sigma_z^{(1)} \sigma_z^{(2)} - \sigma_z^{(1)} \sigma_z^{(2)} \sigma_z^{(2)} = \sigma_z^{(1)} - \sigma_z^{(1)} = 0. \quad (5)$$

Therefore $[\sigma_z^{(1)} \pm \sigma_z^{(2)}, \Pi_z] = 0$, and by (3) we have

$$[\hat{C}_\pm, \Pi_z] = 0. \quad (6)$$

A jump produced by $\hat{C}_\pm$ preserves the z -parity: if a pre-jump pure state $|\psi\rangle$ is a parity eigenstate, so is the post-jump normalized state : $|\psi'\rangle = \hat{C}_\pm |\psi\rangle / \|\hat{C}_\pm |\psi\rangle\|$.

2.2 First jump from $|++\rangle$: projection to definite parity subspace

For pure-state jumps we use the standard update rule

$$p_k = \langle \psi | C_k^\dagger C_k | \psi \rangle dt, \quad (7)$$

with $k \in \{+, -\}$ and dt a short time step.

It is convenient to evaluate the *direction* of the unnormalized state $\hat{C}_\pm |++\rangle$ using the numerators of (3):

$$(\sigma_z^{(1)} + \sigma_z^{(2)}) |++\rangle = 2(|11\rangle - |00\rangle) = -2\sqrt{2} |\phi_-\rangle, \quad (8)$$

$$(\sigma_z^{(1)} - \sigma_z^{(2)}) |++\rangle = 2(|10\rangle - |01\rangle) = -2\sqrt{2} |\psi_-\rangle. \quad (9)$$

Hence the *normalized* post-jump states are

$$|++\rangle \xrightarrow{C_+} -|\phi_-\rangle, \quad |++\rangle \xrightarrow{C_-} -|\psi_-\rangle. \quad (10)$$

Overall minus signs are global phases and carry no physical effect; we keep them to make the subsequent alternation visually clear.

Equations (10) show that the first click already fixes the z -parity: C_+ lands in the even sector (Bell- ϕ states), while C_- lands in the odd sector (Bell- ψ states).

2.3 Subsequent jumps: two-state cycles within each parity sector

Using again only the numerators (because normalization is restored at each jump), we find the action on Bell states:

$$\hat{C}_+ |\phi_-\rangle = -|\phi_+\rangle, \quad \hat{C}_+ |\phi_+\rangle = -|\phi_-\rangle, \quad (11)$$

$$\hat{C}_- |\psi_-\rangle = -|\psi_+\rangle, \quad \hat{C}_- |\psi_+\rangle = -|\psi_-\rangle. \quad (12)$$

After renormalization, this yields the deterministic cycles

$$-|\phi_-\rangle \xrightarrow{C_+} |\phi_+\rangle \xrightarrow{C_+} -|\phi_-\rangle \xrightarrow{C_+} \dots, \quad -|\psi_-\rangle \xrightarrow{C_-} |\psi_+\rangle \xrightarrow{C_-} -|\psi_-\rangle \xrightarrow{C_-} \dots. \quad (13)$$

Because of $[C_\pm, \Pi_z] = 0$, the “wrong” channel is forbidden inside each parity subspace.

2.4 Jump probabilities at each step

From (3),

$$C_{\pm}^{\dagger} C_{\pm} = \frac{\gamma}{2} (S_z^{(1)} \pm S_z^{(2)})^2 = \frac{\gamma}{2} \left[(S_z^{(1)})^2 + (S_z^{(2)})^2 \pm 2S_z^{(1)} S_z^{(2)} \right]. \quad (14)$$

Using $S_z^2 = \frac{1}{4} \mathbb{I}$ and $S_z^{(1)} S_z^{(2)} = \frac{1}{4} \sigma_z^{(1)} \sigma_z^{(2)} = \frac{1}{4} \Pi_z$, we obtain the identity:

$$C_{\pm}^{\dagger} C_{\pm} = \frac{\gamma}{4} (\mathbb{I} \pm \Pi_z). \quad (15)$$

Therefore, for any pure state $|\psi\rangle$, the transition probability rate is:

$$p_{\pm}(|\psi\rangle) = \langle \psi | C_{\pm}^{\dagger} C_{\pm} | \psi \rangle dt = \frac{\gamma}{4} \left(1 \pm \langle \Pi_z \rangle_{\psi} \right) dt. \quad (16)$$

For the state $|++\rangle$, since $\langle \sigma_z \rangle_{|++\rangle} = 0$, we have $\langle \Pi_z \rangle_{|++\rangle} = \langle \sigma_z \rangle_{|++\rangle} \langle \sigma_z \rangle_{|++\rangle} = 0$, hence

$$p_{+}(|++\rangle) = p_{-}(|++\rangle) = \frac{\gamma}{4} dt. \quad (17)$$

The two channels are equiprobable on the initial state, and the total jump rate is $\frac{\gamma}{2} dt$.

For any even-parity state (eigenvalue +1 of Π_z ; in particular $|\phi_{\pm}\rangle$),

$$p_{+}(\text{even}) = \frac{\gamma}{2} dt, \quad p_{-}(\text{even}) = 0. \quad (18)$$

For any odd-parity state (eigenvalue -1; in particular $|\psi_{\pm}\rangle$),

$$p_{-}(\text{odd}) = \frac{\gamma}{2} dt, \quad p_{+}(\text{odd}) = 0. \quad (19)$$

Thus, once the first click has selected a parity subspace, only the corresponding channel can ever fire; conditioned on a click, the state update within that sector is deterministic as in (13).

2.5 State by state transitions

Let us collect both the post-jump states and the *unconditional* jump probabilities over a time step dt :

Pre-jump state	Channel	Post-jump state & probability
$ ++\rangle$	C_{+}	$- \phi_{-}\rangle, p_{+} = \frac{\gamma}{4} dt$
$ ++\rangle$	C_{-}	$- \psi_{-}\rangle, p_{-} = \frac{\gamma}{4} dt$
$- \phi_{-}\rangle$	C_{+}	$ \phi_{+}\rangle, p_{+} = \frac{\gamma}{2} dt$
$ \phi_{+}\rangle$	C_{+}	$- \phi_{-}\rangle, p_{+} = \frac{\gamma}{2} dt$
(even sector)	C_{-}	forbidden, $p_{-} = 0$
$- \psi_{-}\rangle$	C_{-}	$ \psi_{+}\rangle, p_{-} = \frac{\gamma}{2} dt$
$ \psi_{+}\rangle$	C_{-}	$- \psi_{-}\rangle, p_{-} = \frac{\gamma}{2} dt$
(odd sector)	C_{+}	forbidden, $p_{+} = 0$

The selection rule is a direct consequence of $[C_{\pm}, \Pi_z] = 0$: jumps cannot mix parity sectors. Equation (15) makes this quantitative.

2.6 Variances within each channel's cycle

For Bell states the correlation matrix is diagonal:

$$\langle \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)} \rangle = \begin{cases} \text{diag}(+1, -1, +1) & \text{for } |\phi_+\rangle \text{ on } (\alpha = x, y, z), \\ \text{diag}(-1, +1, +1) & \text{for } |\phi_-\rangle, \\ \text{diag}(+1, +1, -1) & \text{for } |\psi_+\rangle, \\ \text{diag}(-1, -1, -1) & \text{for } |\psi_-\rangle. \end{cases}$$

Using

$$\hat{J}_\alpha^2 = \frac{1}{2} (\hat{\mathbb{I}} + \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)}), \quad \langle \hat{J}_\alpha \rangle = 0 \text{ for Bell states,}$$

one gets

$$\Delta J_\alpha^2 = \langle \hat{J}_\alpha^2 \rangle = \frac{1}{2} (1 + \langle \hat{\sigma}_\alpha^{(1)} \otimes \hat{\sigma}_\alpha^{(2)} \rangle).$$

So we summarize in a table of variances ΔJ_α^2 , for the two cycles):

state	ΔJ_x^2	ΔJ_y^2	ΔJ_z^2
$ \phi_-\rangle$	0	1	1
$ \phi_+\rangle$	1	0	1
$ \psi_-\rangle$	0	0	0
$ \psi_+\rangle$	1	1	0

3 Ensemble-averaged concurrence over trajectories

For a pure two-qubit state $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, the concurrence is $\mathcal{C}(|\psi\rangle) = 2|ad - bc|$. For a separable state it is 0 and for any Bell states it is 1. Hence $\mathcal{C}(|++\rangle) = 0$ and $\mathcal{C}(|\phi_\pm\rangle) = \mathcal{C}(|\psi_\pm\rangle) = 1$.

Along each trajectory the concurrence is a step process: it remains 0 until the *first* click, and jumps to 1 thereafter (all subsequent clicks keep the state maximally entangled within the same parity subspace).

The waiting time to the first click is exponential with rate

$$\lambda = \text{Tr} \left[(\hat{C}_+^\dagger \hat{C}_+ + \hat{C}_-^\dagger \hat{C}_-) \cdot \rho_{++} \right] = \frac{\gamma}{2},$$

so the survival probability of no click up to time t is $P_0(t) = e^{-\lambda t} = e^{-(\gamma/2)t}$. Hence the ensemble-averaged concurrence over many trajectories is

$$\boxed{\bar{\mathcal{C}}(t) = \text{prob}(\text{at least one click by } t) = 1 - e^{-(\gamma/2)t}}$$

With $\gamma = 1$, $\bar{\mathcal{C}}(t) = 1 - e^{-t/2}$.

4 Spin squeezing: Kitagawa-Ueda vs Wineland

Define collective spins (for $N = 2$)

$$\hat{J}_\alpha = \frac{1}{2} (\hat{\sigma}_\alpha^{(1)} + \hat{\sigma}_\alpha^{(2)}), \quad \alpha \in \{x, y, z\}.$$

Kitagawa–Ueda (KU).

$$\xi_S^2 \equiv \frac{4}{N} \min_{\mathbf{n} \perp \langle \hat{\mathbf{J}} \rangle} \Delta J_{\mathbf{n}}^2, \quad \Delta J_{\mathbf{n}}^2 = \langle \hat{J}_{\mathbf{n}}^2 \rangle - \langle \hat{J}_{\mathbf{n}} \rangle^2.$$

For the Bell states, $\langle \hat{\mathbf{J}} \rangle = \mathbf{0}$ and at least one orthogonal direction has zero variance. Therefore

$$\boxed{\xi_S^2(|\phi_{\pm}\rangle) = \xi_S^2(|\psi_{\pm}\rangle) = 0}$$

Before the first click the state is $|++\rangle$, a CSS state aligned along x , with $\min_{\mathbf{n} \perp \hat{x}}(\Delta J_{\mathbf{n}}^2) = \frac{N}{4} = \frac{1}{2}$, hence $\xi_S^2(|++\rangle) = 1$ as expected. Averaging over many trajectories,

$$\boxed{\overline{\xi_S^2}(t) = P_0(t) \cdot 1 + (1 - P_0(t)) \cdot 0 = e^{-(\gamma/2)t}}$$

Wineland (phase-sensitivity) parameter.

$$\xi_R^2 \equiv \frac{\min_{\mathbf{n} \perp \langle \hat{\mathbf{J}} \rangle} \Delta J_{\mathbf{n}}^2}{\|\langle \hat{\mathbf{J}} \rangle\|^2}.$$

All Bell states satisfy $\|\langle \hat{\mathbf{J}} \rangle\| = 0$; thus ξ_R^2 is undefined since it diverges). In this parity-selected dynamics the KU definition is the meaningful one, whereas the Wineland parameter cannot be used because the mean spin vanishes.

5 Adding efficiency η

5.1 Observed and unobserved part

Assume identical detection efficiencies on the two outputs, $\eta_1 = \eta_2 \equiv \eta \in [0, 1]$. The physical jump operators are unchanged,

$$\hat{C}_{\pm} = \sqrt{\gamma} \frac{\hat{S}_z^{(1)} \pm \hat{S}_z^{(2)}}{\sqrt{2}} = \sqrt{\gamma} \frac{\hat{\sigma}_z^{(1)} \pm \hat{\sigma}_z^{(2)}}{2\sqrt{2}}, \quad \hat{\Pi}_z = \hat{\sigma}_z^{(1)} \otimes \hat{\sigma}_z^{(2)},$$

and the unconditional master equation (average evolution) is unaffected by η . At the level of the unravelling, each physical channel splits into an *observed* and an *unobserved* part:

$$\text{observed: } \sqrt{\eta} \hat{C}_{\pm}, \quad \text{unobserved: } \sqrt{1 - \eta} \hat{C}_{\pm}.$$

5.2 Jump probabilities as post-jump norms (observed vs physical)

Using $\hat{C}_{\pm}^{\dagger} \hat{C}_{\pm} = \frac{\gamma}{4} (\hat{\mathbb{I}} \pm \hat{\Pi}_z)$, the physical and observed click probabilities over dt are

$$p_{\pm}^{\text{phys}}(\psi) = \langle \psi | \hat{C}_{\pm}^{\dagger} \hat{C}_{\pm} | \psi \rangle dt = \frac{\gamma}{4} (1 \pm \langle \hat{\Pi}_z \rangle_{\psi}) dt, \quad p_{\pm}^{\text{obs}}(\psi) = \eta p_{\pm}^{\text{phys}}(\psi).$$

From $|++\rangle$ one has $\langle \hat{\Pi}_z \rangle = 0$ and therefore $p_{+}^{\text{obs}} = p_{-}^{\text{obs}} = \frac{\eta\gamma}{4} dt$, total observed rate $\eta\gamma/2$. Because $[\hat{C}_{\pm}, \hat{\Pi}_z] = 0$, inside the even (odd) subspace the $-$ ($+$) channel remains strictly forbidden even for $\eta < 1$.

5.3 Concurrence with finite efficiency

For a pure two-qubit state $|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle$, $\mathcal{C}(|\psi\rangle) = 2|ad - bc|$; thus $\mathcal{C}(|++\rangle) = 0$ and $\mathcal{C}(\text{Bell}) = 1$. Along each trajectory the first jump makes $\mathcal{C}$ jump from 0 to 1 and it stays there; averaging over physical trajectories it gives

$$\overline{\mathcal{C}}_{\text{phys}}(t) = 1 - e^{-(\gamma/2)t} \quad (\text{independent of } \eta).$$

For the conditional estimate (based on the observed record), between two observed clicks the state in each subspace is a mixture of two orthogonal Bell states with weights p and $1 - p$. For such a Bell-diagonal form one has

$$\mathcal{C}_{\text{cond}}(t) = |2p(t) - 1| = e^{-(1-\eta)\gamma(t-t_0)}.$$

Observed clicks arrive with rate $\lambda = \eta \cdot \gamma/2$ and reset $\mathcal{C}_{\text{cond}}$ to 1. In stationary conditions the time since the last observed click, τ , is exponentially distributed with density $\lambda e^{-\lambda\tau}$; therefore

$$\langle \mathcal{C}_{\text{cond}} \rangle_{\text{ss}} = \int_0^\infty e^{-(1-\eta)\gamma\tau} \lambda e^{-\lambda\tau} d\tau = \frac{\lambda}{\lambda + (1-\eta)\gamma} = \boxed{\frac{\eta}{2-\eta}}.$$

With $\gamma = 1$, $\lambda = \eta/2$ and the same result follows.

6 Spin squeezing with finite efficiency

Let's discuss how it Kitagawa-Ueda spin squeezing behaves in the finite efficiency case. By using the results before, we show that in the even sector, using $(\Delta J_x^2, \Delta J_y^2, \Delta J_z^2) = (0, 1, 1)$ for $|\phi_-\rangle$ and $(1, 0, 1)$ for $|\phi_+\rangle$, the mixture $\rho_{\text{even}}(t) = p|\phi_{\text{start}}\rangle\langle\phi_{\text{start}}| + (1-p)|\phi_{\text{other}}\rangle\langle\phi_{\text{other}}|$ has

$$\Delta J_x^2 = p, \quad \Delta J_y^2 = 1 - p, \quad \Delta J_z^2 = 1,$$

so that $\min_{\mathbf{n}} \Delta J_{\mathbf{n}}^2 = \min\{p, 1-p\} = \frac{1}{2}(1 - |2p - 1|)$. Therefore

$$\xi_{S,\text{even}}^2(t) = 1 - |2p(t) - 1| = 1 - e^{-(1-\eta)\gamma(t-t_0)}.$$

Every observed + click resets $\xi_{S,\text{even}}^2$ to 0; between clicks it increases towards 1. Averaging over the exponential distribution of τ with rate $\lambda = \eta\gamma/2$ yields

$$\langle \xi_{S,\text{even}}^2 \rangle_{\text{ss}} = \int_0^\infty (1 - e^{-(1-\eta)\gamma\tau}) \lambda e^{-\lambda\tau} d\tau = \boxed{1 - \frac{\eta}{2-\eta} = \frac{2(1-\eta)}{2-\eta}}.$$

In the odd sector, using $(\Delta J_x^2, \Delta J_y^2, \Delta J_z^2) = (0, 0, 0)$ for $|\psi_-\rangle$ and $(1, 1, 0)$ for $|\psi_+\rangle$, any mixture $p|\psi_-\rangle\langle\psi_-| + (1-p)|\psi_+\rangle\langle\psi_+|$ has $\min_{\mathbf{n}} \Delta J_{\mathbf{n}}^2 = 0$, hence

$$\xi_{S,\text{odd}}^2(t) = 0 \quad \text{for all } t, \quad \langle \xi_{S,\text{odd}}^2 \rangle_{\text{ss}} = 0.$$

Finally, starting from $|++\rangle$, the first (physical) click selects the parity sector with probability 1/2 each. Averaging the stationary values over sectors gives the overall stationary mean

$$\boxed{\langle \xi_S^2 \rangle_{\text{ss, overall}} = \frac{1}{2} \left(1 - \frac{\eta}{2-\eta} \right) + \frac{1}{2} \cdot 0 = \frac{1-\eta}{2-\eta}}.$$

7 Generalized CSS initial state ($\phi = 0$)

We now take as initial state a two-spin coherent spin state (CSS) with polar angle θ and $\phi = 0$:

$$|\psi(\theta)\rangle = |n\rangle^{\otimes 2}, \quad |n\rangle = \sin \frac{\theta}{2} |1\rangle + \cos \frac{\theta}{2} |0\rangle.$$

The jump operators remain

$$\hat{C}_{\pm} = \sqrt{\gamma} \frac{\hat{S}_z^{(1)} \pm \hat{S}_z^{(2)}}{\sqrt{2}} = \sqrt{\frac{\gamma}{8}} (\hat{\sigma}_z^{(1)} \pm \hat{\sigma}_z^{(2)}), \quad \hat{\Pi}_z = \hat{\sigma}_z^{(1)} \otimes \hat{\sigma}_z^{(2)},$$

with the convention $\hat{\sigma}_z |1\rangle = +|1\rangle$, $\hat{\sigma}_z |0\rangle = -|0\rangle$.

Jump probabilities from the post-jump norm

Using $\hat{C}_{\pm}^{\dagger} \hat{C}_{\pm} = \frac{\gamma}{4} (\hat{\mathbb{I}} \pm \hat{\Pi}_z)$,

$$p_{\pm}(\theta) = \langle \psi(\theta) | \hat{C}_{\pm}^{\dagger} \hat{C}_{\pm} | \psi(\theta) \rangle = \frac{\gamma}{4} \left(1 \pm \langle \hat{\Pi}_z \rangle_{\psi(\theta)} \right).$$

Since $|\psi(\theta)\rangle = |n\rangle \otimes |n\rangle$ with $\langle \hat{\sigma}_z \rangle_{|n\rangle} = -\cos \theta$,

$$\langle \hat{\Pi}_z \rangle_{\psi(\theta)} = \langle \hat{\sigma}_z \rangle_{|n\rangle}^2 = \cos^2 \theta.$$

Hence

$$\boxed{p_+(\theta) = \frac{\gamma}{4} (1 + \cos^2 \theta), \quad p_-(\theta) = \frac{\gamma}{4} \sin^2 \theta, \quad p_+ + p_- = \frac{\gamma}{2}.}$$

Normalized post-jump states

Expanding $|\psi(\theta)\rangle$ in the computational basis and acting with $\hat{\sigma}_z^{(1)} \pm \hat{\sigma}_z^{(2)}$, one finds (up to the overall factor $\sqrt{\gamma/8}$)

$$\begin{aligned} (\hat{\sigma}_z^{(1)} + \hat{\sigma}_z^{(2)}) |\psi(\theta)\rangle &= 2 \sin^2 \frac{\theta}{2} |11\rangle - 2 \cos^2 \frac{\theta}{2} |00\rangle, \\ (\hat{\sigma}_z^{(1)} - \hat{\sigma}_z^{(2)}) |\psi(\theta)\rangle &= 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} (|10\rangle - |01\rangle). \end{aligned}$$

Therefore

$$|\psi'_+\rangle = \frac{\sin^2(\frac{\theta}{2}) |11\rangle - \cos^2(\frac{\theta}{2}) |00\rangle}{\sqrt{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})}}, \quad |\psi'_-\rangle = \frac{|10\rangle - |01\rangle}{\sqrt{2}} = |\psi_-\rangle.$$

Concurrence after one observed click and channel-averaged value

For a pure state $|\Psi\rangle = a |00\rangle + b |01\rangle + c |10\rangle + d |11\rangle$, $\mathcal{C}(|\Psi\rangle) = 2|ad - bc|$. Thus

$$\mathcal{C}_-(\theta) = 1, \quad \mathcal{C}_+(\theta) = \frac{2 \sin^2(\frac{\theta}{2}) \cos^2(\frac{\theta}{2})}{\sin^4(\frac{\theta}{2}) + \cos^4(\frac{\theta}{2})} = \frac{\sin^2 \theta}{2 - \sin^2 \theta}.$$

Averaging over the two channels with weights $p_{\pm}/(p_+ + p_-)$ gives

$$\boxed{\bar{\mathcal{C}}(\theta) = \frac{p_+ \mathcal{C}_+ + p_- \mathcal{C}_-}{p_+ + p_-} = \frac{1}{2} \left[\sin^2 \theta + (1 + \cos^2 \theta) \frac{\sin^2 \theta}{2 - \sin^2 \theta} \right].}$$

Checks: $\bar{\mathcal{C}}(0) = \bar{\mathcal{C}}(\pi) = 0$, $\bar{\mathcal{C}}(\frac{\pi}{2}) = 1$.

Kitagawa–Ueda spin squeezing after one observed click

For $N = 2$, $\xi_S^2 = \frac{4}{N} \min_{\mathbf{n} \perp \langle \hat{\mathbf{J}} \rangle} \Delta J_{\mathbf{n}}^2$, with $\hat{J}_\alpha = \frac{1}{2}(\hat{\sigma}_\alpha^{(1)} + \hat{\sigma}_\alpha^{(2)})$. In the $-$ channel the post-jump state is the singlet $|\psi_-\rangle$, for which $\Delta J_\alpha^2 = 0$ for all α and $\xi_{S,-}^2 = 0$.

In the $+$ channel, for $|\psi'_+\rangle = a|00\rangle + d|11\rangle$ (real a, d) one has $\langle \hat{J}_x \rangle = \langle \hat{J}_y \rangle = 0$, $\langle \hat{J}_z \rangle = |d|^2 - |a|^2$, and

$$\Delta J_x^2 = \frac{1}{2}(1 + \langle \hat{\sigma}_x \otimes \hat{\sigma}_x \rangle), \quad \Delta J_y^2 = \frac{1}{2}(1 + \langle \hat{\sigma}_y \otimes \hat{\sigma}_y \rangle).$$

A short calculation yields

$$\langle \hat{\sigma}_x \otimes \hat{\sigma}_x \rangle = -\frac{\sin^2 \theta}{2 - \sin^2 \theta}, \quad \langle \hat{\sigma}_y \otimes \hat{\sigma}_y \rangle = +\frac{\sin^2 \theta}{2 - \sin^2 \theta},$$

so that

$$\Delta J_x^2 = \frac{\cos^2 \theta}{2 - \sin^2 \theta}, \quad \Delta J_y^2 = \frac{1}{2 - \sin^2 \theta}.$$

Since $\langle \hat{\mathbf{J}} \rangle$ is along z , the minimum transverse variance is $\min(\Delta J_x^2, \Delta J_y^2) = \Delta J_x^2$. With $N = 2$,

$$\boxed{\xi_{S,+}^2(\theta) = 2 \Delta J_x^2 = \frac{2 \cos^2 \theta}{2 - \sin^2 \theta}.$$

Therefore the channel-averaged KU parameter is

$$\boxed{\overline{\xi_S^2}(\theta) = \frac{p_+ \xi_{S,+}^2 + p_- \xi_{S,-}^2}{p_+ + p_-} = \frac{1}{2}(1 + \cos^2 \theta) \frac{2 \cos^2 \theta}{2 - \sin^2 \theta} = \frac{(1 + \cos^2 \theta) \cos^2 \theta}{2 - \sin^2 \theta}.$$

Edge cases: $\overline{\xi_S^2}(0) = \overline{\xi_S^2}(\pi) = 1$ (product states, no squeezing); $\overline{\xi_S^2}(\frac{\pi}{2}) = 0$ (both channels yield perfectly number-squeezed Bell states).

In plots we set $\gamma = 1$ (dimensionless rates); the formulas above are independent of γ once probabilities are normalized by $p_+ + p_- = \gamma/2$.